Data Structures

Fall 2025

11.
Doubly Linked List
Circular Linked List
Circular Doubly

Introduction – Doubly LL

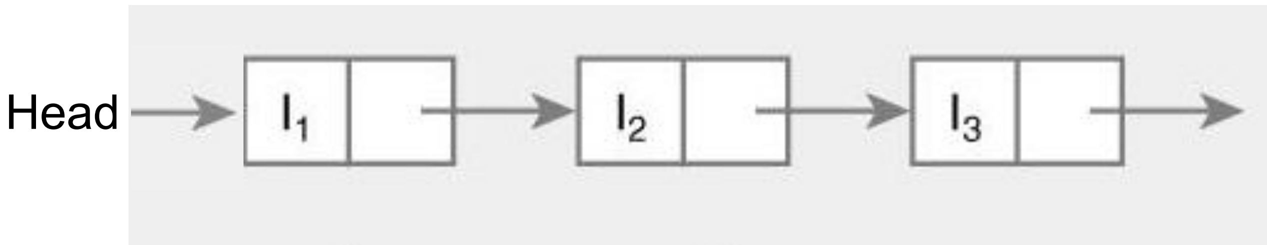
- The singly linked list contains only one pointer field i.e. every node holds an address of next node.
- The singly linked list is uni-directional i.e. we can only move from one node to its successor.
- This limitation can be overcome by **Doubly linked list.**

Doubly Linked List

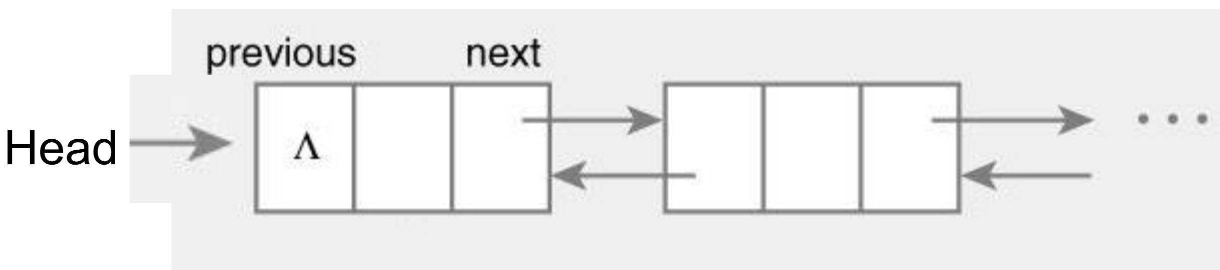
- In Doubly linked list, each node has two pointers.
- One pointer to its successor (NULL if there is none) and one pointer to its predecessor (NULL if there is none).
- These pointers enable bi-directional traversing.

Comparison of Linked List

A Singly Linked List



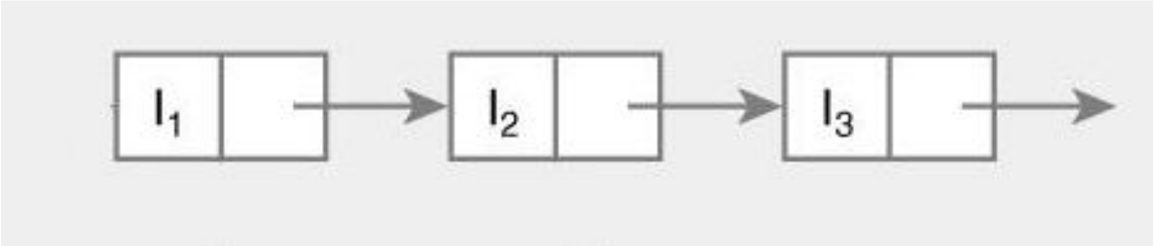
A Doubly Linked List



Comparison of Linked List

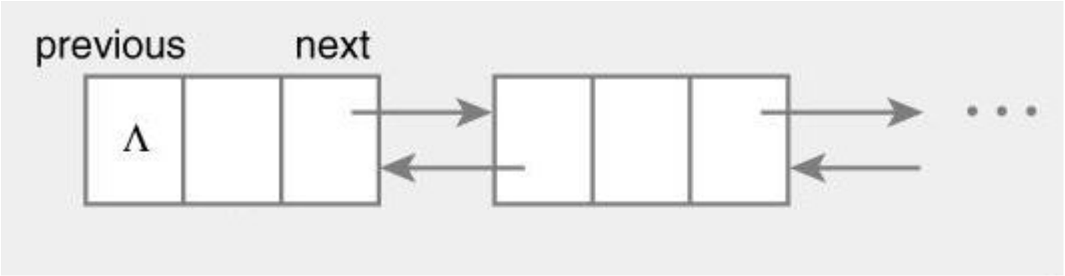
- **Linked list**

```
struct Node {
    int data;
    Node* next;
};
```



- **Doubly linked list**

```
struct Node {
    Node *previous;
    int data;
    Node *next;
};
```



Linked List Class

```
class list
{
    private:
        node *head,*tail;

    public:
        list()
        {
            head=NULL;
            tail=NULL;
        }
```

Insertion

- In insertion process, element can be inserted in three different places
  - At the beginning of the list
  - At the end of the list
  - At the specified position.

- To insert a node in doubly linked list, you must update pointers in both predecessor and successor nodes.

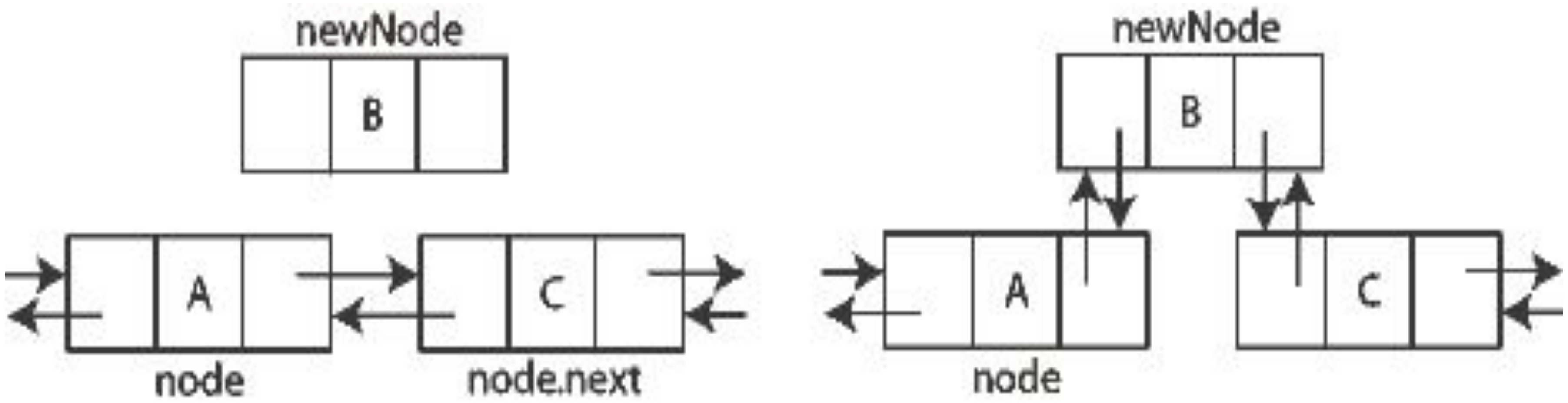
Insertion at head

```
void insert_at_head(int v)
{
    node *temp=new node;
    temp->data=v;
    temp->next=NULL;
    temp->prev=NULL;

    if(head==NULL&tail==NULL)
    {
        head=temp;
        tail=temp;
    }
    else
    {
        temp->next=head;
        head->prev=temp;
        head=temp;
    }
}
```

Insertion at end

```
void insert_at_end(int v)
{
    node *temp=new node;
    temp->data=v;
    temp->next=NULL;
    temp->prev=NULL;

    if(head==NULL&tail==NULL)
    {
        head=temp;
        tail=temp;
    }
    else
    {
        tail->next=temp;
        temp->prev=tail;
        tail=temp;
    }
}
```

Insertion at location



Insertion at location

```
void insert_at_pos(int loc,int v)
{
    node *temp=new node;
    temp->data=v;
    temp->next=NULL;
    temp->prev=NULL;

    if(head==NULL&tail==NULL)
    {
        head=temp;
        tail=temp;
    }
    else
    {
        node *p=head,*c=head;
        for(int i=1;i<loc;i++)
        {
            p=c;
            c=c->next;
        }
        p->next=temp;
        temp->prev=p;
        c->prev=temp;
        temp->next=c;
    }
}
```

Deletion

- In deletion process, element can be deleted from three different places
  - From the beginning of the list
  - From the end of the list
  - From the specified position in the list.

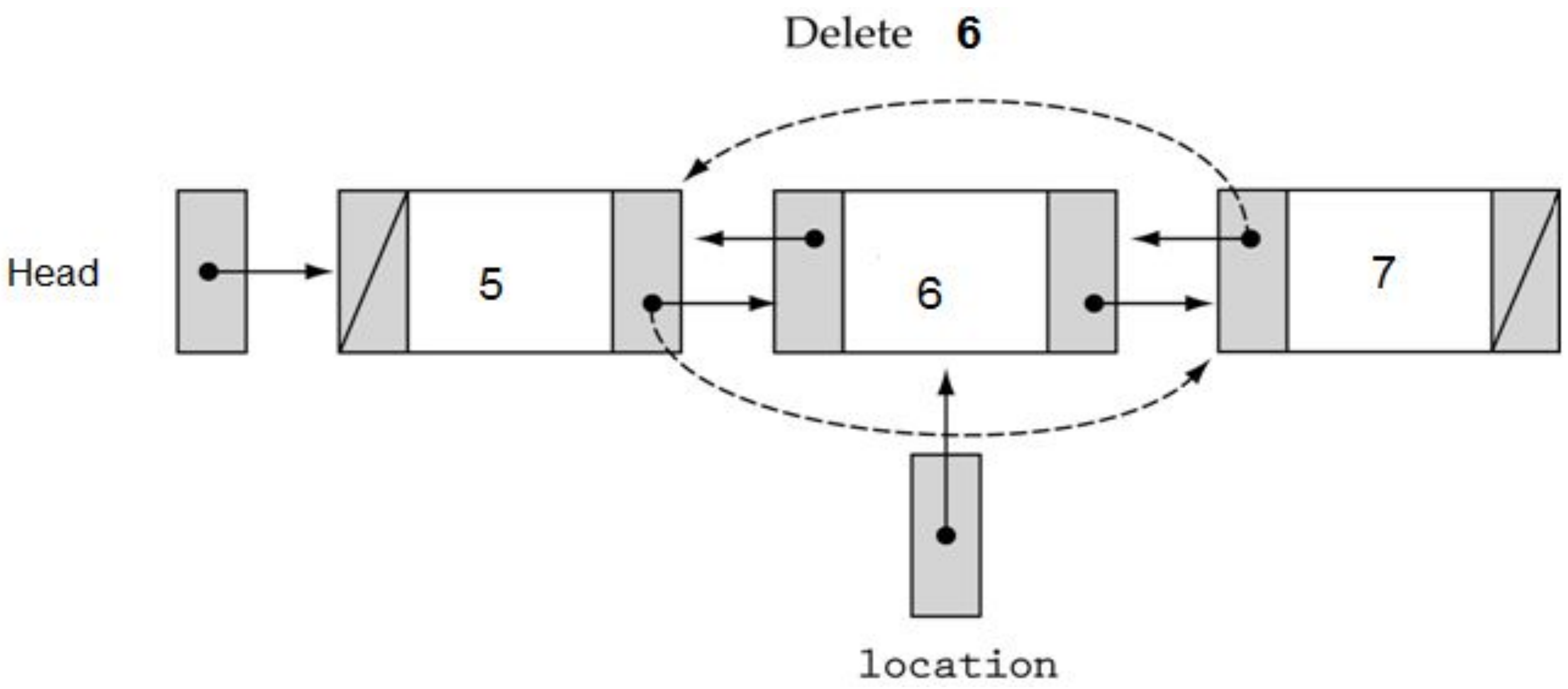
- When the node is deleted, the memory allocated to that node is released and the previous and next nodes of that node are linked

Deletion at head and tail

```
void delete_at_head()
{
    node *q=head;
    head=head->next;
    head->prev=NULL;
    q->next=NULL;
    delete q;
    q=NULL;
}
```

```
void delete_at_tail()
{
    node *q=tail;
    tail=tail->prev;
    tail->next=NULL;
    q->prev=NULL;
    delete q;
    q=NULL;
}
```

Deletion at location



Deletion at location

```
void delete_at_loc(int l)
{
    node *pre=head,*curr=head;
    for(int i=1;i<l;i++)
    {
        pre=curr;
        curr=curr->next;
    }
    pre->next=curr->next;
    curr->next->prev=pre;
    delete curr;
    curr=NULL;
}
```

Printing List

```
void display()
{
    cout<<"\nfrom head to tail"<<endl;
    node *p=head;
    while(p!=NULL)
    {
        cout<<p->data<<"\t";
        p=p->next;
    }
}
```

```
void reverse_display()
{
    cout<<"\nfrom tail to head"<<endl;
    node *q=tail;
    while(q!=NULL)
    {
        cout<<q->data<<"\t";
        q=q->prev;
    }
}
```

Advantages

- The doubly linked list is bi-directional, i.e. it can be traversed in both backward and forward direction.
- The operations such as insertion, deletion and searching can be done from both ends.
- Predecessor and successor of any element can be searched quickly

Disadvantages

- It consume more memory space.
- There is a large pointer adjustment during insertion and deletion of element.
- It consumes more time for few basic list operations.

Introduction – Circular LL

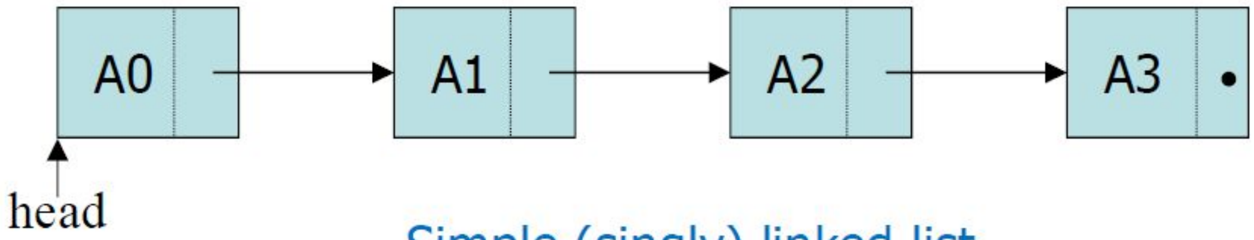
- Just like singly linked list contains only one pointer field i.e. every node holds an address of next node.

- The singly linked list is uni-directional i.e. we can only move from one node to its successor. Circular can be singly or doubly circular

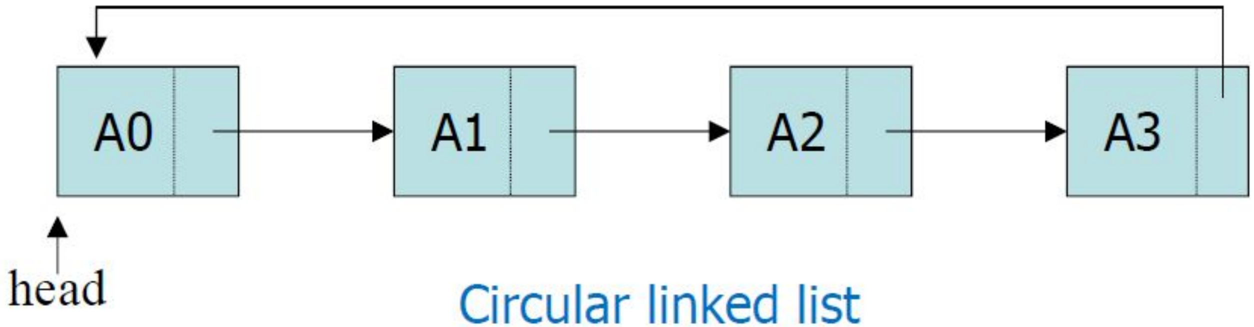
- The last node is connected to first

Comparison of Linked List

- A linked list in which the last node points to the first node



Simple (singly) linked list

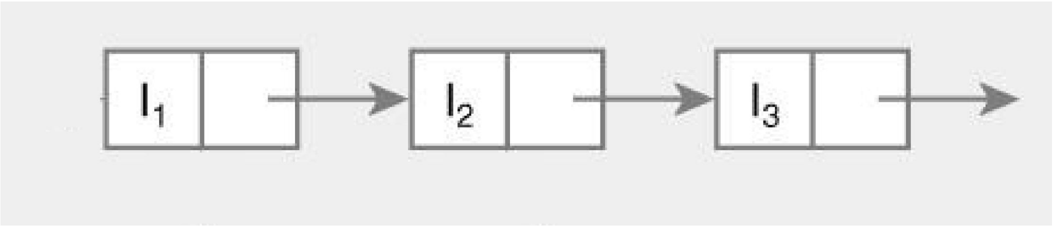


Circular linked list

Comparison of Linked List

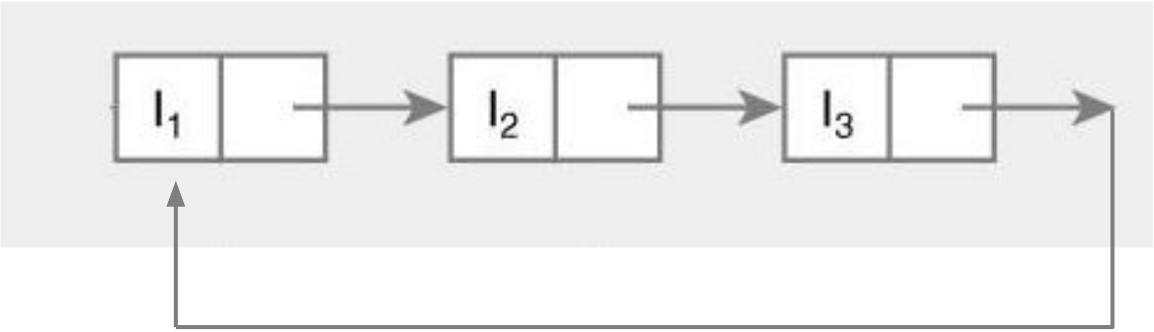
- **Linked list**

```
struct Node {
    int data;
    Node* next;
};
```



- **Circular linked list**

```
struct Node {
    int data;
    Node *next;
};
```



Linked List Class

Singly Circular

```
struct node{
    int data;
    node *next;
};

class clist{
    private:
        node *tail;

    public:
        clist()
        {
            tail=NULL;
        }
```

Doubly Circular

```
struct node{
    node *prev;
    int data;
    node *next;
};

class dclist{
    private:
        node *tail;

    public:
        dclist()
        {
            tail=NULL;
        }
```

Insertion

- In insertion process, element can be inserted in three different places
  - At the beginning of the list
  - At the end of the list
  - At the specified position.

Insertion at head

Singly Circular

```
void add_to_head(int v)
{
    node *temp=new node;
    temp->data=v;
    temp->next=NULL;

    if(tail==NULL)
    {
        tail=temp;
        tail->next=temp;
    }
    else
    {
        temp->next=tail->next;
        tail->next=temp;
    }
}
```

Doubly Circular

```
void add_to_head(int v)
{
    node *temp=new node;
    temp->data=v;
    temp->prev=NULL;
    temp->next=NULL;

    if (tail==NULL) {
        tail=temp;
        tail->prev=temp;
        tail->next=temp;
    }
    else {
        temp->next=tail->next;
        temp->prev=tail;
         tail->next->prev=temp;
        tail->next=temp;
    }
}
```

Insertion at tail

Singly Circular

```
void add_to_tail(int v)
{
    node *temp=new node;
    temp->data=v;
    temp->next=NULL;

    if(tail==NULL)
    {
        tail=temp;
        tail->next=temp;
    }
    else
    {
        temp->next=tail->next;
        tail->next=temp;
        tail=temp;
    }
}
```

Doubly Circular

```
void add_to_tail(int v)
{
  node *temp=new node;
  temp->data=v;
  temp->prev=NULL;
  temp->next=NULL;

  if (tail==NULL) {
    tail=temp;
    tail->prev=temp;
    tail->next=temp;
  }
  else {
    temp->next=tail->next;
    temp->prev=tail;
     tail->next->prev=temp;
    tail->next=temp;
     tail=temp;
  }
}
```

Insertion at location

Singly Circular

```
void add_to_loc(int l,int v)
{
    node *temp=new node;
temp->data=v;
temp->next=NULL;

if(tail==NULL)
{
    tail=temp;
    tail->next=temp;
}
else
{
node *pre=tail,*curr=tail;
for(int i=1;i<=l;i++)
    {
        pre=curr;
        curr=curr->next;
    }
    pre->next=temp;
    temp->next=curr;
}
```

Doubly Circular

```
void add_to_loc(int l, int v)
{
  node *temp=new node;
  temp->data=v;
  if (tail==NULL) {
    tail=temp;
    tail->prev=temp;
    tail->next=temp;
  }
  else {
    node *pre=tail,*curr=tail;
    for(int i=1;i<=l;i++) {
      prev=curr;
      curr=curr->next;
    }
    prev->next=temp;
    temp->next=curr;
    curr->prev=temp;
    temp->prev=prev;
  }
}
```

Deletion

- In deletion process, element can be deleted from three different places
  - From the beginning of the list
  - From the end of the list
  - From the specified position in the list.
- When the node is deleted, the memory allocated to that node is released and the previous and next nodes of that node are linked

Deletion at head and end

Singly Circular

```
void delete_at_start()
{
    node *q=tail->next;
    tail->next=q->next;
    delete q;
    q=NULL;
}

void delete_at_end()
{
    node *q=tail;
    node *temp=tail;
    while(temp->next!=tail)
    {
        temp=temp->next;
    }
    tail=temp;
    tail->next=q->next;
}
```

Doubly Circular

```
void delete_at_start()
{
    node *q=tail->next;
    tail->next=q->next;
    q->next->prev=q->prev;
    delete q;
}

void delete_at_end()
{
    tail=tail->prev;
    node *q=tail->next;
    tail->next=q->next;
    q->next->prev=q->prev;
    delete q;
}
```

Display Linked List

Singly Circular

```
void display()
{
    node *p=tail->next;
do
{
        cout<<p->data<<"\t";
        p=p->next;
        }while(p!=tail->next);
}
```

Doubly Circular

```
void display()
{
   node *p=tail->next;
   do
   {
      cout<<p->data<<“\t”;
      p=p->next;
   } while(p!=tail->next);
}

void display_reverse()
{
   node *p=tail;
   do
   {
      cout<<p->data<<“\t”;
      p=p->prev;
   } while(p!=tail);
}
```

Advantages

- Whole list can be traversed by starting from any point
  - Any node can be starting point
  - What is the stopping condition?
- Fewer special cases to consider during implementation
  - All nodes have a node before and after it
- Used in the implementation of other data structures
  - Circular linked lists are used to create circular queues
  - Circular doubly linked lists are used for implementing Fibonacci heaps

Disadvantages

- Finding end of list and loop control is harder
  - No NULL to mark beginning and end

Any Question So Far?